AALBORG UNIVERSITY
Department of Computer Science

Software Engineering for AI Systems
Spring 2027

Portfolio 1: Software Architecture

Case: Short Anonymised Case Name

Student Name

October 27, 2026

Assessment criteria (delete this box before submitting):

- Scenarios are measurable and traceable to the business goal.
- Diagrams are consistent with the scenarios and the ADRs.
- ADRs name real alternatives and the trade-off that decided them.
- Debt risks and the AI Act classification are specific to the case, not a generic list.

1 Problem and Context

State the business goal, the data available, and why ML is needed rather than fixed rules. Describe the system at the level of components and data types, and anonymise names where needed.

Stakeholder	Interest in the system
…	…
…	…

Table 1: Stakeholders.

2 Quality-Attribute Scenarios

Give 3–5 scenarios in six-part form [1]. Say which qualities matter most, which conflict, and what a wrong prediction costs.

At least one scenario is a *model-quality scenario*, e.g. recall on a time-based hold-out set, with the cost of a missed case stated.

ID	Source	Stimulus	Environment	Artefact	Response	Measure
QA-1	…	…	…	…	…	…
QA-2	…	…	…	…	…	…
QA-3	…	…	…	…	…	…

Table 2: Quality-attribute scenarios.

3 Architecture

Show the system as C4 context and container diagrams, with the AI and non-AI parts marked. Name the existing systems it must integrate with (e.g. ERP, historian, PLC/SCADA, data platform) and the network zones it crosses.

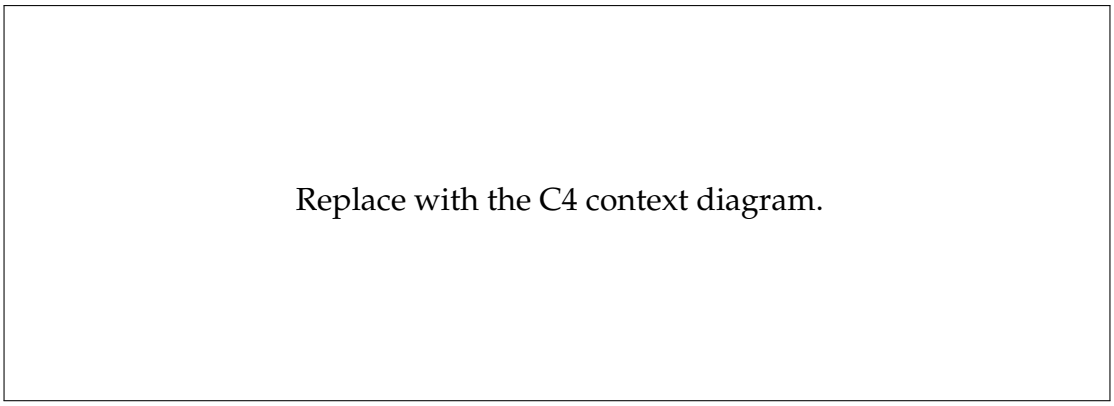


Figure 1: System context.

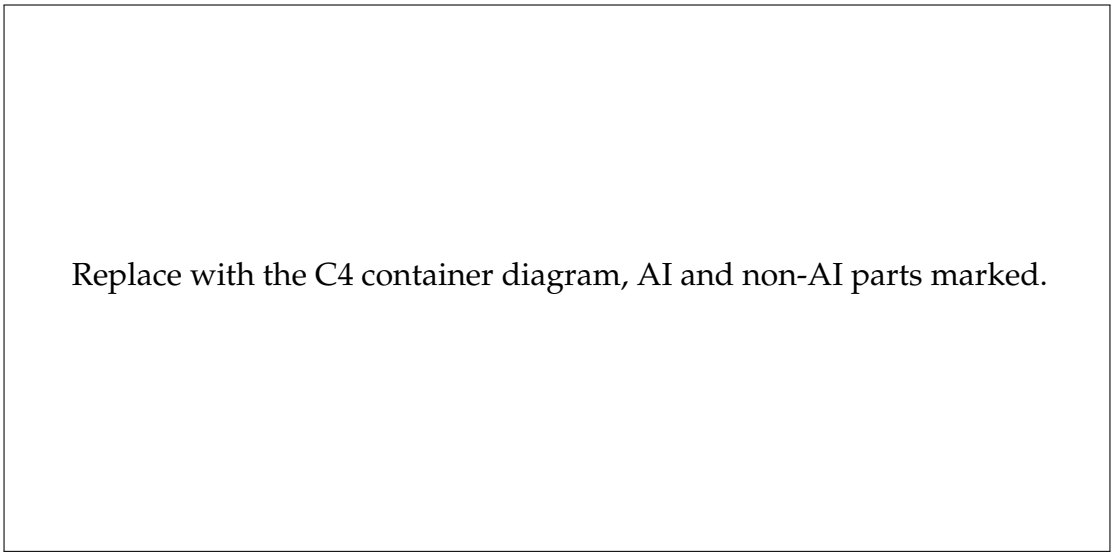


Figure 2: Containers.

4 Architecture Decisions

Record at least two decisions. Each ADR names the scenarios from Table 2 that drove it.

4.1 ADR-1: Title of the Decision

Status: Proposed / accepted.

Context: The problem and the scenarios (QA-. . .) that drive it.

Alternatives: At least two real alternatives.

Decision: The chosen alternative.

Consequences: The trade-off that decided it, and what gets harder.

4.2 ADR-2: Title of the Decision

Status: . . .

Context: ...

Alternatives: ...

Decision: ...

Consequences: ...

5 EU AI Act Classification

State the risk category of the system under the AI Act [2], justify it with reference to the relevant article or annex, and say what follows from it for the design.

6 Technical-Debt Risks

Name the debt the design will accumulate [3], and how the design limits it.

Debt risk	Where in the system	How the design limits it
...	...	...
...	...	...

Table 3: Technical-debt risks.

7 LLM Considerations

Only if your case uses an LLM or another foundation model [4]: how prompts and model versions are managed, how outputs are evaluated, and which guardrails apply. Otherwise delete this section.

8 AI Usage

Briefly describe your use of AI tools on this assignment:

- *Code generation:* ...
- *Grammar and writing:* ...
- *Clarification of technical aspects:* ...

Bibliography

[1] L. Bass, P. Clements, and R. Kazman, *Software Architecture in Practice, 4th Edition*, 1st ed. Addison-Wesley Professional, 2021, oCLC: 1251808773.

[2] European Parliament and Council of the European Union, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)," Official Journal of the European Union, L series, Jul. 2024.

[3] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison, "Hidden Technical Debt in Machine Learning Systems," in *Advances in Neural Information Processing Systems 28 (NeurIPS 2015)*, 2015, pp. 2503–2511.

[4] L. Bass, Q. Lu, I. Weber, and L. Zhu, *Engineering AI systems*. Boston, MA: Addison Wesley, Mar. 2025.